

CS & IT ENGINEERING

Data Structure

Stack & Queues

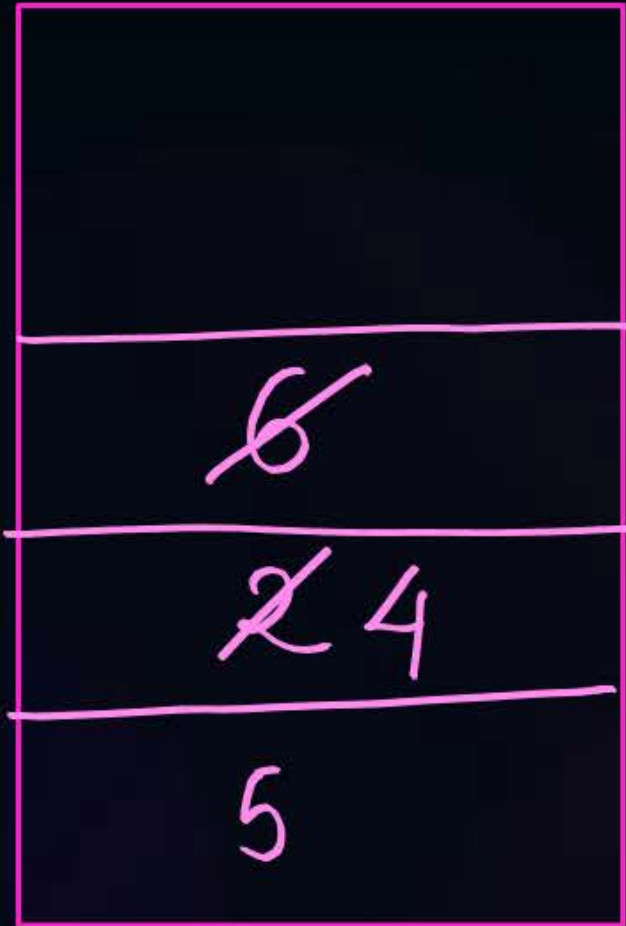
DPP.- 01

Discussion Notes



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#Q. Consider the following sequence of operations on an empty stack:
push (5); push (2); pop(); push(4); push(6); p=pop(); q=pop(); r=pop();
The value of $p + q - r$ is- (5).



$$p + q - r$$

$$6 + 4 - 5 = (5)$$

$$p = \text{pop}()$$

$$p = 6$$

$$q = \text{pop}() = 4$$

$$r = \text{pop} = 5$$

#Q. Which of the following includes the applications of stack?

A Recursive function calls ✓ Recursion Runtime

B HTML and XML Tag matching ✓ $\langle \text{Head} \rangle \langle / \text{Head} \rangle$ stack

C ✓ Checking if an expression contains balanced parantheses. $\langle \text{Body} \rangle \langle / \text{Body} \rangle$

D Finding the maximum element in a given sequence.

Max



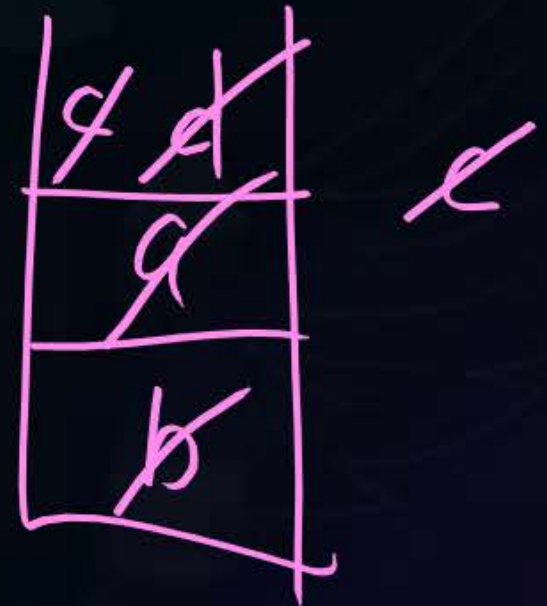
[MCQ]

#Q. A stack is implemented using array. S represents the pointer to the top element in the stack. Initially the stack contains the elements: $a(\text{top})$, b . Assume $\text{Push}(S, i)$ push an element i into the stack at index S . Whenever a Push operation will be performed, it will returns $S++$ after the push operation. $\text{Pop}()$ pops the topmost element and returns the next top index. $\text{Top}()$ is a function that returns the topmost element of the stack. Consider the following statements:

P: $\text{Top}(\text{Pop}(\text{Pop}(\text{Pop}(\text{Push}(\text{Push}(S, c), d)))) = a$

Q: $\text{Pop}(\text{Pop}(\text{Pop}(\text{Pop}(\text{Push}(\text{Pop}(\text{Push}(S, c)), d)))) \neq a$

Which of the following statements is/are INVALID?



A

P only

B

Q only

C

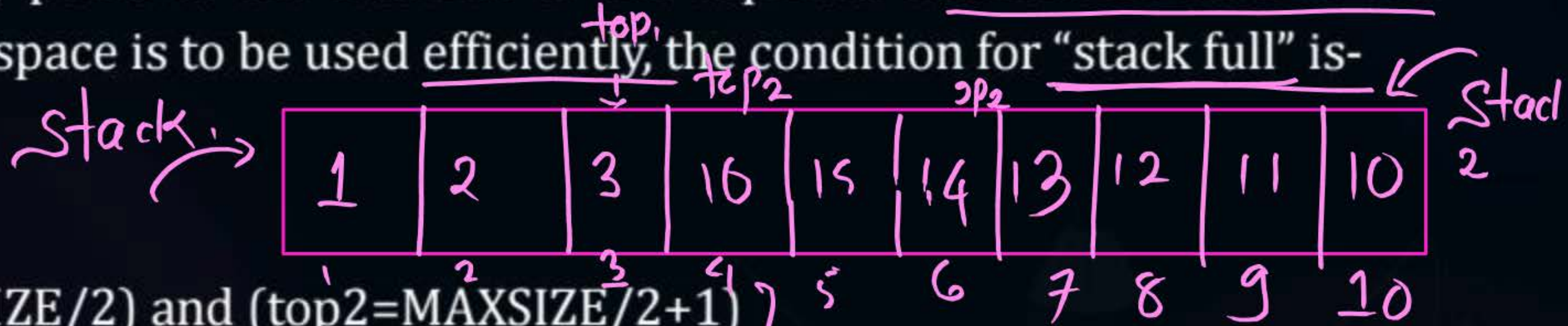
Both P and Q ✓

D

Neither P nor Q

[MCQ]

#Q. A single array $A[1 \dots \text{MAXSIZE}]$ is used to implement two stacks. The two stacks grow from opposite ends of the array. Variables top1 and top2 ($\text{top1} < \text{top2}$) point to the location of the topmost element in each of the stacks. If the space is to be used efficiently, the condition for "stack full" is-



A

$(\text{top1} = \text{MAXSIZE}/2)$ and $(\text{top2} = \text{MAXSIZE}/2 + 1)$

B

$(\text{top1} = \text{MAXSIZE}/2)$ or $(\text{top2} = \text{MAXSIZE}/2 + 1)$

C

$\text{top1} + \text{top2} = \text{MAXSIZE}$

D

$\text{top1} = \text{top2} - 1$

$$\text{top}_1 = \text{top}_2 - 1$$

[D]

#Q. The attribute of three arithmetic operators in some programming language are given below.

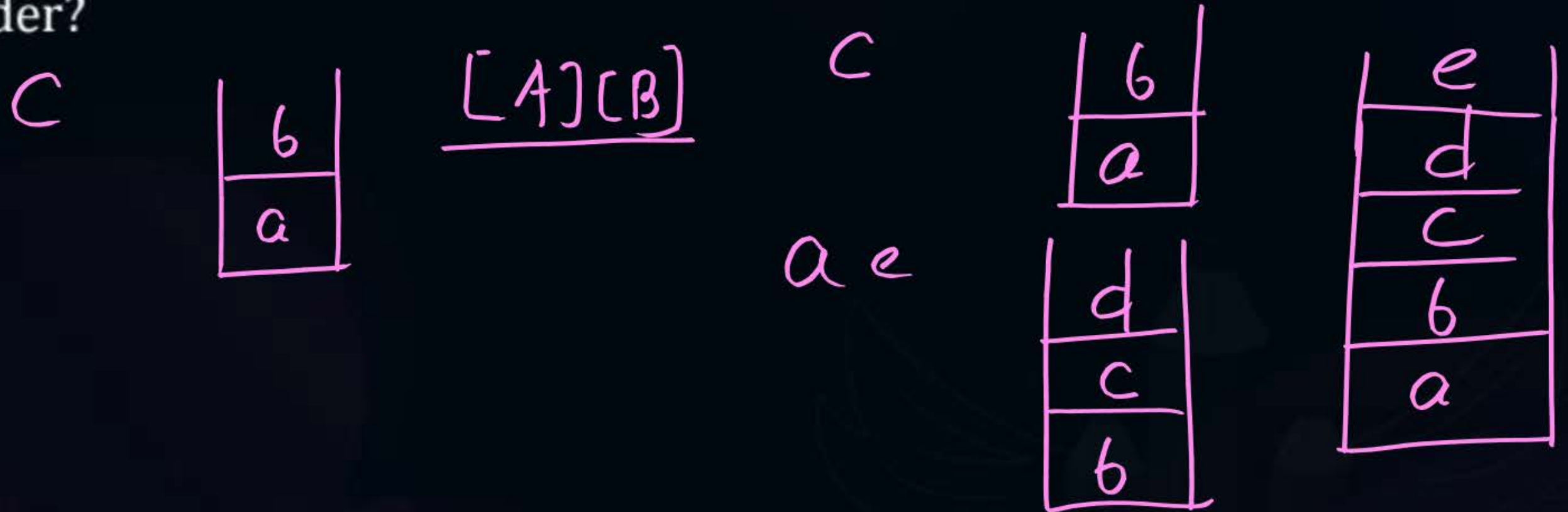
OPERATOR	PRECEDENCE	ASSOCIATIVITY	ARITY
+	High	Left ✓	Binary ✓
-	Medium ✓	Right ✓	Binary ✓
*	Low	Left	Binary

The value of the expression $4 - (6 + 2) - 8 * 2$ in this language is 8.

$$\begin{aligned} & 4 - (8 - 8) * 2 \\ & (4 - 0) * 2 = 4 * 2 \end{aligned}$$

[MSQ]

#Q. Which one of the following permutations cannot be obtained in the output string using a stack and assuming that the input sequence is a, b, c, d, e in the same order?



A c d e a b ~~✗~~

B a e b c d ~~✗~~

C c d e b a

D e d c b a

[MCQ]

#Q. A stack is implemented using array of size 4. S represents the pointer to the top element in the stack. Initially the stack contains the elements- a(top), b. Assume $\text{Push}(S, i)$ push an element i into the stack at index S. Whenever a Push operation will be performed, it will returns $S++$ after the push operation. $\text{Pop}()$ pops the topmost element and returns the next top index. $\text{isEmpty}()$ returns TRUE if the stack is empty. $\text{isFull}()$ returns TRUE if the stack is full. Consider the following statements:

P: $\text{isFull}(\text{Push}(\text{Pop}(\text{Push}(\text{Push}(S, c), d))), e) = \text{TRUE}$ ✓
 Q: $\text{isEmpty}(\text{Push}(\text{Pop}(\text{Pop}(\text{Push}(\text{Pop}(\text{Push}(S, c)), d)))), e) = \text{FALSE}$

Which of the following statements is/are VALID? (3)

[C]

A

P only

B

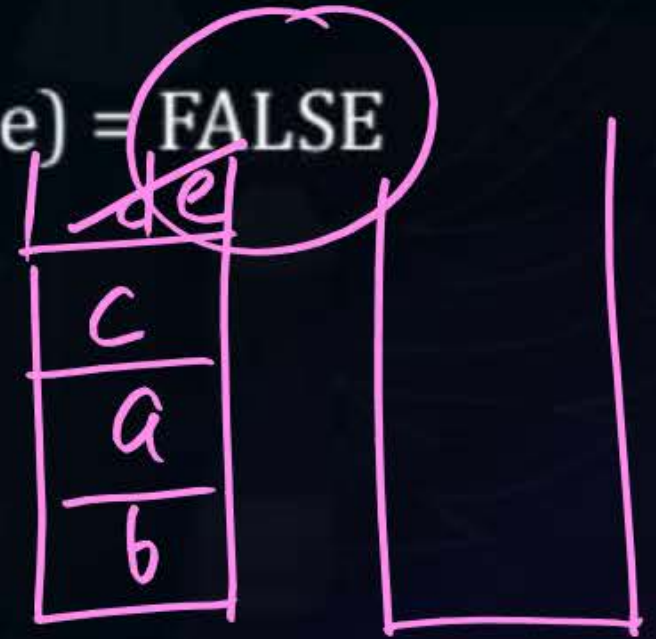
Q only

C

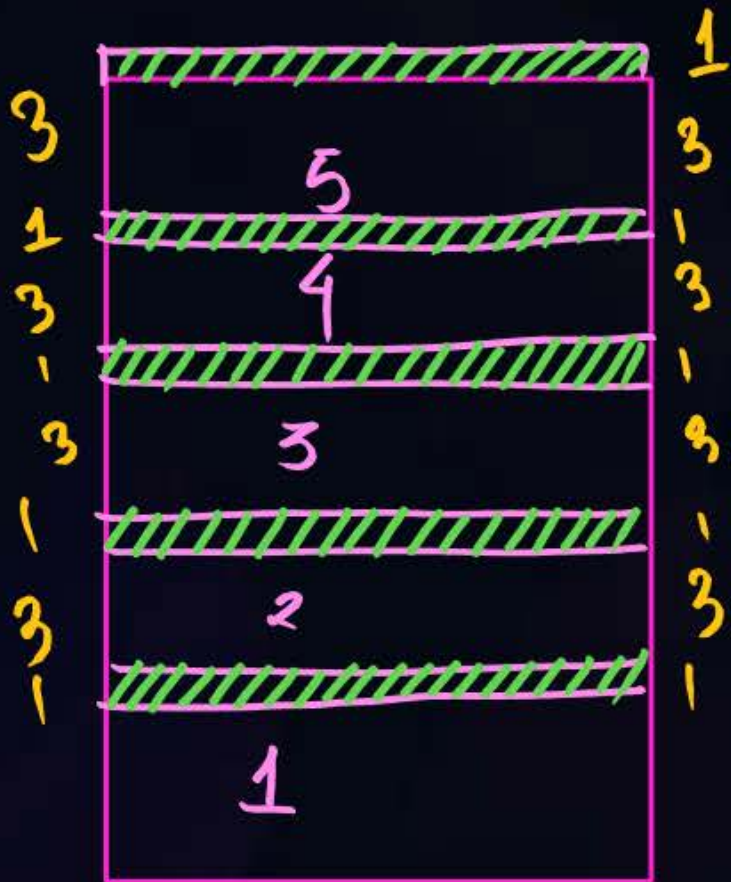
Both P and Q ✓

D

Neither P nor Q



- #Q. Let S be a stack of size $n \geq 1$. Starting with the empty stack, suppose we push the first 5 natural numbers in sequence, and then perform 5 pop operations. Assume that Push and Pop operations take 3 seconds each, and 1 seconds elapse between the end of one such stack operation and the start of the next operation. The average stack-life of an element of this stack is 17.



Stack life

$$5 - 1 \text{ sec}$$

$$4 = 3 + 6 = 9$$

$$3 = 5 + 12 = 17$$

$$2 = 7 + 18 = 25$$

$$1 = 9 + 24 = 33$$

33

25

17

9

25

85/5 = 17



THANK - YOU